WORK ORDER 001

Make a fitted tray

Design a tray for your pocket notebook. A partner will build it from your measurements and instructions.

The job: make a shallow paper tray that sits on a desk. Your notebook must lie flat inside and come out easily. You will design one tray and build your partner's tray.

Requirement	Starting design
Clearance around the notebook	1/8 inch at EACH of the four sides
Wall height	3/4 inch on all four walls
Materials	Loose paper, scissors, tape, pencil, straightedge
Tools and record	Your tape measure and pocket notebook

1 Measure the real notebook

Close your notebook. Measure the outside length and width of its cover. Keep the tape straight and use the same endpoints when checking. Ask your partner to remeasure. If readings disagree, compare the setup and check again. **Write the chosen measurements and units in your notebook.**

2 Calculate the base and the paper blank

The **base** is the flat inside bottom. The **blank** is the larger rectangle of paper before you fold the walls. Show the additions in your notebook.

Base length = notebook length + 1/8 + 1/8
Base width = notebook width + 1/8 + 1/8
Blank length = base length + 3/4 + 3/4
Blank width = base width + 3/4 + 3/4
All measurements above are in inches.

3 Send a work order your partner can use

Make a small labelled sketch and write these on a piece of scrap paper:
Owner: ___ **Builder:** ___
Base: ___ in long by ___ in wide
Blank: ___ in long by ___ in wide
Walls: 3/4 in high
Label the base, walls, cuts, and folds on the sketch. Mark your small sketch **NTS** (not to scale). Keep a copy of the dimensions in your notebook.

Before the swap: check your arithmetic with the teacher. Make sure the blank fits the available sheet in one orientation. Ask for a plan if it does not fit. Exchange work orders; ask about unclear dimensions before cutting.

Build from the work order

Use loose paper for the full-size pattern. Keep your notebook intact.

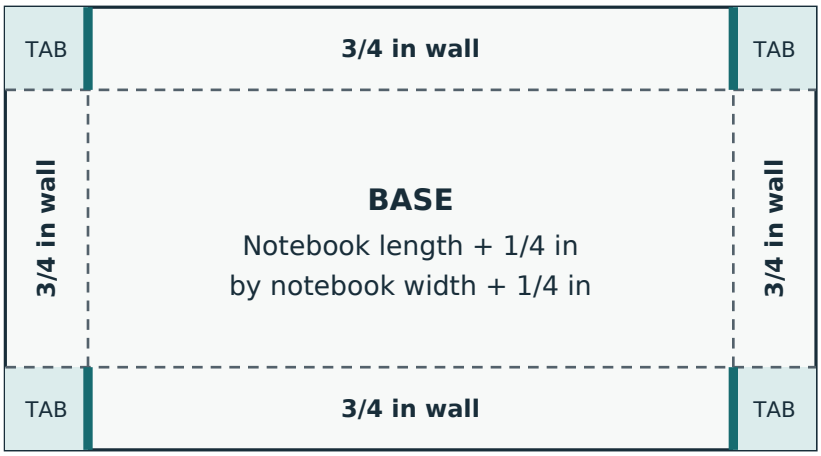

Diagram is NTS. Draw your own full-size pattern from your calculated dimensions.

1. Measure and draw the outer rectangle using the **blank** dimensions. Cut it out.
2. Measure 3/4 inch inward from each edge. Draw all four fold lines straight.
3. Make the four short cuts shown above. Stop at the first fold line. **Keep all four tabs.**
4. Fold the walls up. Turn the tabs inward against the neighboring walls and tape the corners.
5. Label the owner and builder. Return the tray to the owner for inspection.

Test and record

Set the tray flat on the desk. Insert and remove the notebook **three times**. Do not force it. Does it lie flat? Do the corners hold? Does the tray match the work order? Record what happened in your notebook. Point to the relevant dimension or fold.

Revise: correct one problem and repeat the test. If the fit already works, keep its dimensions and improve one unclear label or instruction. Write: **“I changed ___ because the check showed ___.”**

Your notebook spread
Left page: job/date, notebook sketch, measured length and width, units.
Right page: base and blank calculations, fit-test result, revision.
Use short labels and clear numbers. Keep the first attempt readable.

Finish on your own

1. A new notebook is **3 7/8 in wide**. With the same clearances and wall height, find the **base width** and **blank width**. Show your additions.
2. Someone adds only one 3/4-inch wall to the base width. How much too narrow is the blank?
3. Name one result from your real test and the change or check it led to.